

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: E & ECE

2. Subject Name: VLSI Engg. [EC31004] L-T-P: 3-1-0 Credits: 4

3. Pre-requisites: Semiconductor Devices (EC21205), Analog Electronic Circuits (EC21207), Digital Electronic Circuits (EC21202)

4. Syllabus and reference books:

Syllabus:

Introduction to VLSI circuits, Integrated circuit (IC), MOSFET fabrication technology, Basic semiconductor process steps (wafer growth, oxidation, ion-implantation, metallization, etching, CMP etc.), Self-aligned LDD MOSFETs, Layout and stick diagram, Layout rules, Parasitic aware VLSI circuit layout, MOSFET scaling and Technology node, Long and short channel MOSFETs (equations: drain current, subthreshold conduction, threshold voltage), SPICE Model of MOSFETs, BSIM Model, Capacitances in a MOSFET.

Integration of active and passive (resistors, capacitors and inductors) components, Design rules for passive components, Impact of multi-finger MOSFET geometry, Dummy elements in ICs, Guard rings, Modern FETs (FinFETs, gate all around FETs), Interconnect delay in a IC, Packaging technology and Testing, IC Yield statistics.

Introduction to CMOS technology, CMOS inverter, Noise Margin, Voltage-transfer characteristics, Delay in a CMOS inverter circuit, Ring Oscillator, Power consumption in CMOS inverter, Switching activity, Latch-up in CMOS circuits, ESD, other CMOS logic gates, Logical effort, Elmore delay model, Dynamic CMOS Logic, Pass Transistor and Transmission gate Logics, Implementation of memory elements using transmission gates, BiCMOS Logic, Low power VLSI circuits.

Reference Books:

- 1) J. M. Rabaey, A. Chandrakasan and Borivoje Nikolic, "Digital Integrated Circuits – A Design Perspective", Second Edition, Pearson
- 2) S. M. Kang and Y. Leblebici, "CMOS Digital Integrated Circuits – Analysis and Design", Third Edition, McGraw-Hill
- 3) N.H.E. Weste and D. M. Harris, "CMOS VLSI Design – A Circuits and Systems Perspective", Fourth Edition, Pearson
- 4) R. J. Baker, H. W. Li and D. E. Boyce, "CMOS Circuit Design, Layout and Simulation", Prentice-Hall India
- 5) Stephen A. Campbell, "The Science and Engineering of Microelectronic Fabrication", Oxford University Press

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction to VLSI circuits, Integrated circuit (IC), MOSFET scaling and Technology node, Long and short channel MOSFETs (equations: drain current, subthreshold conduction, threshold voltage), SPICE Model of MOSFETs, BSIM Model	6
2.	MOSFET fabrication technology, Basic semiconductor process steps (wafer growth, oxidation, ion-implantation, metallization, etching, CMP etc.), Self-aligned LDD MOSFETs, Modern FETs (FinFETs, gate all around FETs), Layout and stick diagram, Layout rules	6
3.	Capacitances in a MOSFET, Interconnect delay in integrated circuits, Parasitic aware VLSI circuit layout	4
4.	Integration of active and passive (resistors, capacitors and inductors) components, Design rules for passive components, Impact of multi-finger MOSFET geometry, Dummy elements in ICs, Guard rings, ESD	5
5.	Packaging technology and Testing, IC Yield statistics	3
6.	Introduction to CMOS technology, CMOS inverter, Noise Margin, Voltage-transfer characteristics, Delay in a CMOS inverter circuit, Ring Oscillator, Power consumption in CMOS inverter, Switching activity, Latch-up in CMOS circuits	10
7.	Logical effort, Elmore delay model, other CMOS logic gates, Dynamic CMOS Logic, Pass Transistor and Transmission gate Logics, Implementation of memory elements using transmission gates, Low power VLSI circuits	6
8.	BiCMOS Logic	2
Total number of hours		42